Refrigeration

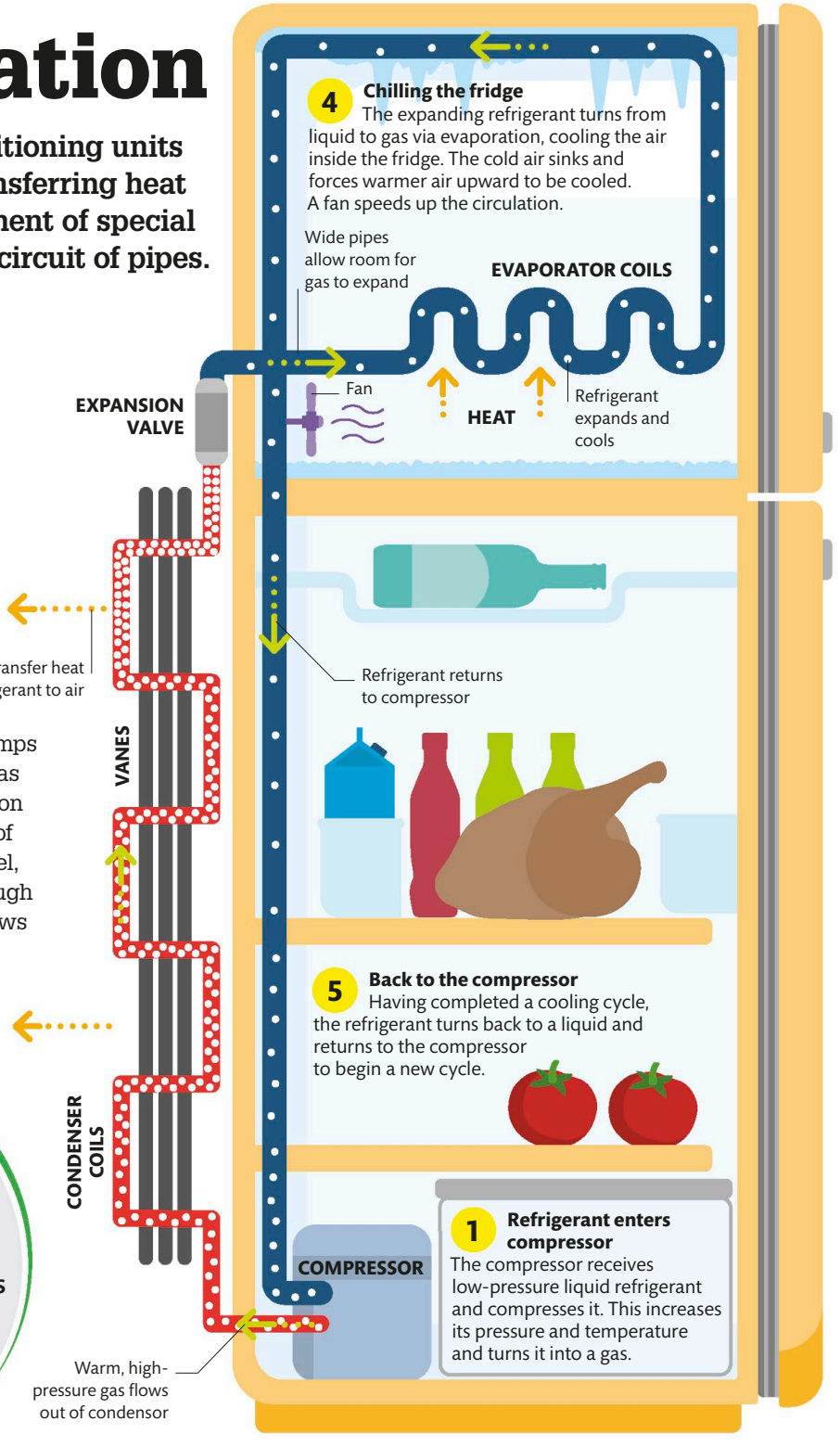
Refrigerators and air-conditioning units cool interior spaces by transferring heat energy through the movement of special chemicals around a coiled circuit of pipes.

3 Refrigerant expands
The liquid flows through an expansion valve that lowers the refrigerant's pressure, making it expand and cool. It passes into the pipes of the evaporator running inside the fridge.

2 Cooling the refrigerant
The gas travels through thin, coiled pipes in the condenser, where metal vanes transfer heat from the refrigerant to the surrounding air. The refrigerant becomes a liquid.

Refrigerators

Refrigerators are effectively heat pumps that move heat energy from cold areas to warm areas—the opposite direction to usual heat flow. A closed system of pipes circulates refrigerant (see panel, opposite), which changes state through compression and expansion and draws heat out of the refrigerator's interior. Freezers work in the same way, just at lower temperatures.



WHAT TEMPERATURE SHOULD A FRIDGE BE SET TO?

A fridge should be kept at about 39°F (4°C). Temperatures above this may not inhibit the growth of bacteria on food.